

Lect 10.1: Differential Equation - Motivation

Thursday, 19 March 2026 1:16 pm



ONCE YOU LEARN THE CONCEPT OF A
DIFFERENTIAL EQUATION, YOU SEE
DIFFERENTIAL EQUATIONS ALL OVER, NO
MATTER WHAT YOU DO.

- GIAN-CARLO ROTA -

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Sophus Lie

Among all of the mathematical
disciplines the theory of
differential equations is the
most important... It furnishes
the explanation of all those
elementary manifestations of
nature which involve time.

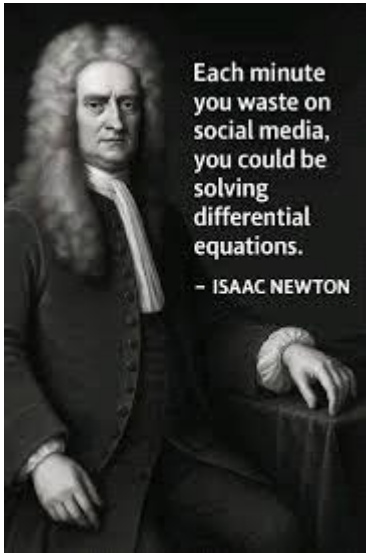
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If you assume continuity, you can open the well-stocked mathematical toolkit of continuous functions and differential equations, the saws and hammers of engineering and physics for the past two centuries (and the foreseeable future).

— Benoit Mandelbrot —

AZ QUOTES



Each minute you waste on social media, you could be solving differential equations.

— ISAAC NEWTON



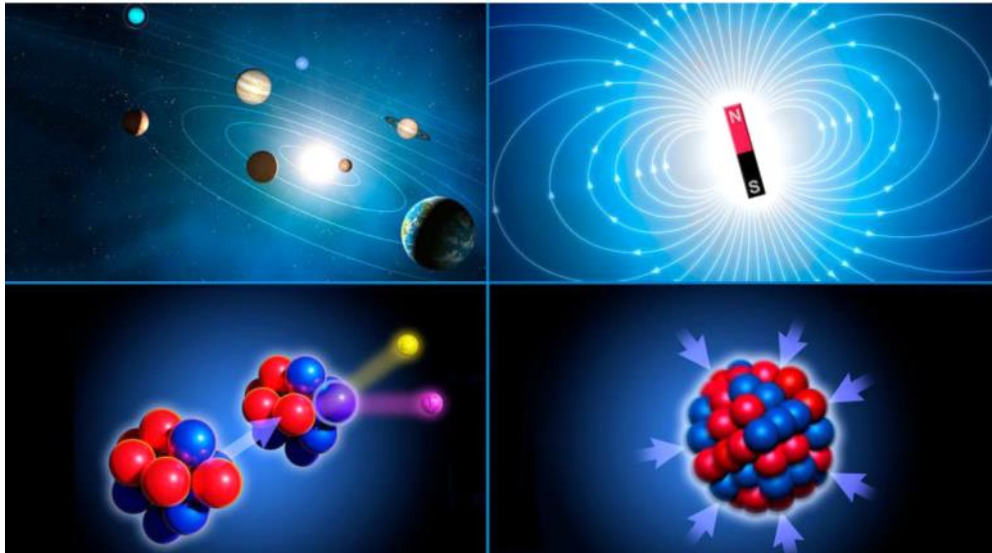
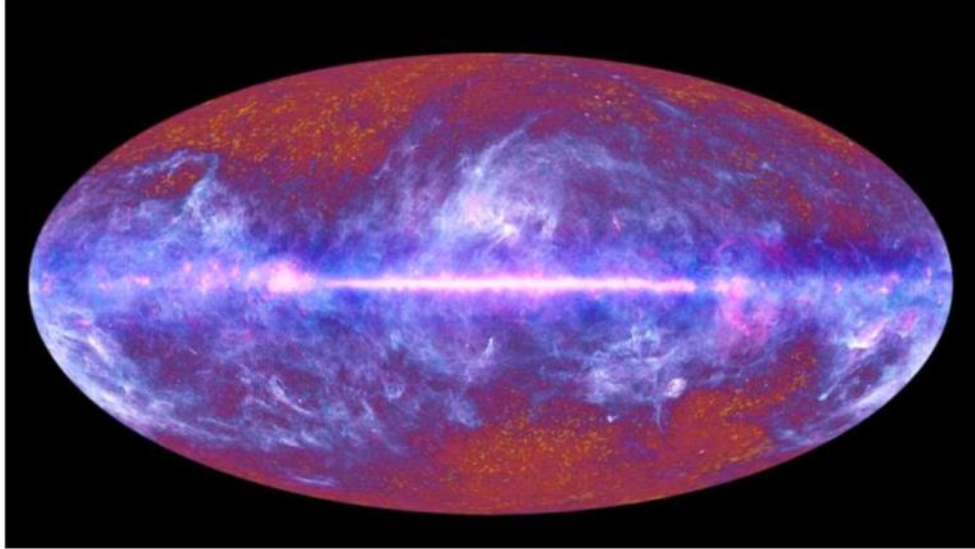
Gottfried Wilhelm Leibniz was the first to state clearly the rules of calculus.



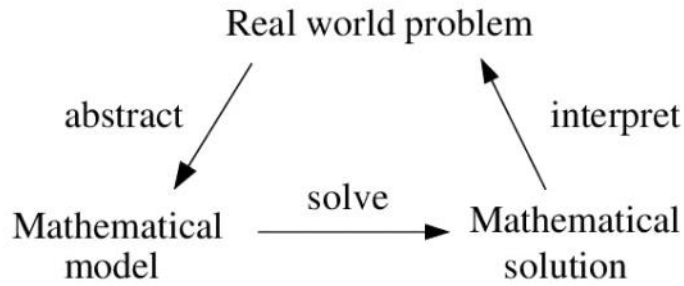
Isaac Newton developed the use of calculus in his laws of motion and universal gravitation.

Why Differential Equations Matters!!!

Universe!!!



Differential Equation



Maxwell's Equations	$\nabla \cdot \mathbf{E} = 0$ $\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{H}}{\partial t}$	$\nabla \cdot \mathbf{H} = 0$ $\nabla \times \mathbf{H} = \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t}$	J.C. Maxwell, 1865
Second Law of Thermodynamics	$dS \geq 0$		L. Boltzmann, 1874
Relativity	$E = mc^2$		Einstein, 1905
Schrodinger's Equation	$i\hbar \frac{\partial}{\partial t} \Psi = H\Psi$		E. Schrodinger, 1927
Information Theory	$H = -\sum p(x) \log p(x)$		C. Shannon, 1949
Chaos Theory	$x_{t+1} = kx_t(1 - x_t)$		Robert May, 1975
Black-Scholes Equation	$\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} - rV = 0$		F. Black, M. Scholes, 1990

Einstein's Field Equation

$$R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

(tells matter/energy
how to curve spacetime)

(tells matter/energy
how to move through
curved spacetime)

SCHRÖDINGER EQUATION

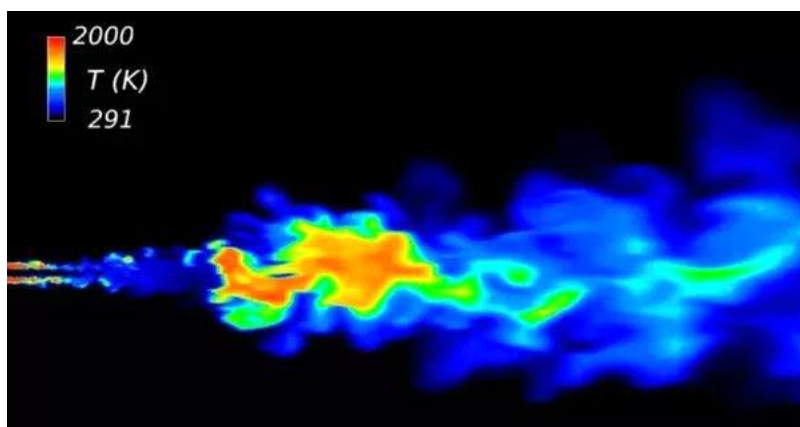
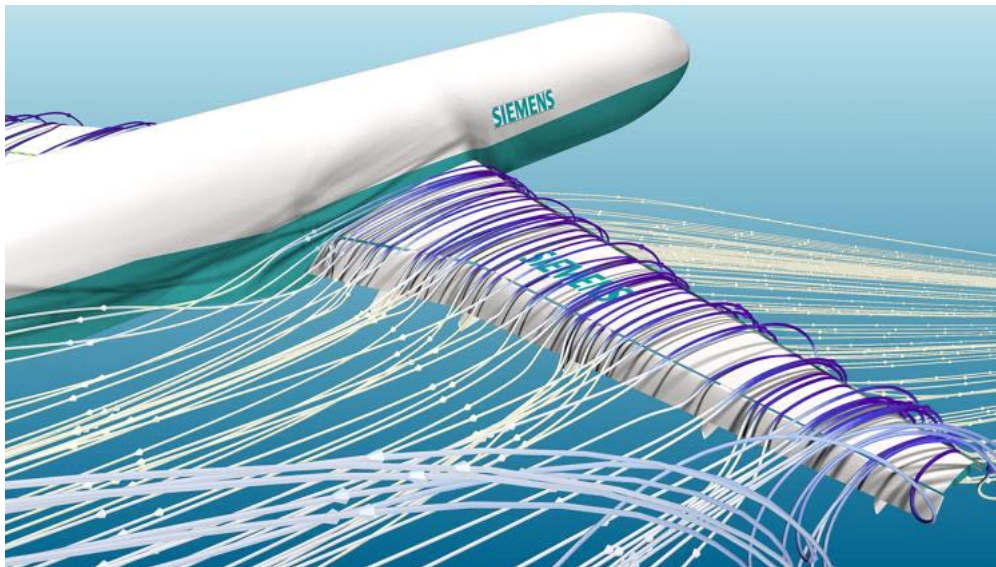
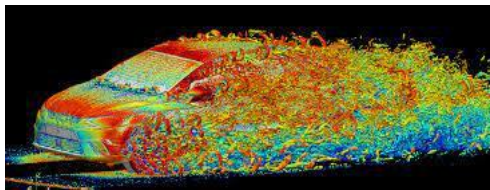
$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

↑

Navier-Stokes Equations

$$\rho \left(\frac{\partial u}{\partial t} + (u \cdot \nabla) u \right) = -\nabla p + \eta \nabla^2 u + f$$

fluid density ρ (mass) $\frac{\partial u}{\partial t}$ (change of velocity over time) $(u \cdot \nabla) u$ (Convection term / inertial term / acceleration / fluid flow velocity / gradient operator) $=$ $-\nabla p$ (fluid pressure / pressure gradient / force) $+\eta \nabla^2 u$ (Diffusion term / viscosity) $+f$ (body forces)



The Navier–Stokes existence and smoothness problem is a \$1 million Millennium Prize challenge from the Clay Mathematics Institute. It asks to prove whether smooth, globally defined, three-dimensional solutions exist for the Navier–Stokes equations (which model fluid flow) or if they break down, failing to describe turbulent behavior at high speeds.  Clay Mathematics Institute +2

